

Quantum KKL for Anticommuting-Pauli Observables

A substantial partial result for the Montanaro–Osborne conjecture

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9 August 2026

Status. Candidate partial result, likely valid. The full quantum L^2 -KKL conjecture remains open.

1 Source problem

Rouzé, Wirth, and Zhang study quantum Talagrand, KKL, and Friedgut inequalities in arXiv:2209.07279, later published in *Communications in Mathematical Physics* [2]. Their Remark 3.10 on page 13 recalls Montanaro and Osborne’s Conjecture 3 from Section 12 of [1]. The source passage is reproduced below.

quantum Boolean function, then

$$\|A - 2^{-n} \text{tr}(A)\|_1 = \|A\|_1 = \|A\|_2^2 = \text{Var}(A) = 1.$$

According to L^1 -Poincaré inequality (3.1),

$$\sum_{j=1}^n \|d_j A\|_1 \geq 1.$$

One may wonder if $\text{Inf}_j^1(A) = \|d_j A\|_1 \approx 1/n$ for all $1 \leq j \leq n$ is possible. However, our Theorem 3.9 for $p = 1$ indicates that this is not the case. There exists j such that $\text{Inf}_j^1(A) \geq C \log(n)/n$ for some $C > 0$.

Remark 3.10. In [MO10a, Conjecture 3 of Section 12], the authors have conjectured a similar KKL-type result for the quantum L^2 -influences $\text{Inf}_j^2(A)$. While this influence coincides with the L^1 -influence $\text{Inf}_j^1(A)$ when A is a classical Boolean function, this is not the case in the quantum setting. Hence, this conjecture in [MO10a] remains open to the best of our knowledge.

Figure 1: Full-width crop of Remark 3.10 on page 13 of the source paper.

Transcription. For a balanced quantum Boolean function $A \in M_2(\mathbb{C})^{\otimes n}$, meaning $A = A^*$, $A^2 = I$, and $\text{tr}(A) = 0$, is there a universal constant $C > 0$ such that

$$\max_{1 \leq j \leq n} \text{Inf}_j^2(A) \geq C \frac{\log n}{n}?$$

Here d_j deletes the Pauli–Fourier terms that are identity on qubit j , and $\text{Inf}_j^2(A) = \|d_j A\|_2^2$ for normalized Schatten norm. The source proves the corresponding result for L^p -influences when $1 \leq p < 2$ and explicitly states that this $p = 2$ endpoint remains open [2]. A later random-restriction treatment also states that the exact conjecture remains open [3].

2 Definitions

Let $\mathcal{P}_n = \{I, X, Y, Z\}^{\otimes n}$ be the Hermitian Pauli strings. With normalized trace 2^{-n}tr , this is an orthonormal basis. For $\Gamma \in \mathcal{P}_n$, write

$$\text{supp}(\Gamma) = \{j : \Gamma_j \neq I\}.$$

For a Pauli expansion $A = \sum_{\Gamma} \hat{A}(\Gamma) \Gamma$, orthogonality gives

$$\text{Inf}_j^2(A) = \sum_{\Gamma: j \in \text{supp}(\Gamma)} |\hat{A}(\Gamma)|^2. \quad (1)$$

We call A an *anticommuting-Pauli observable* if

$$A = \sum_{s=1}^m c_s \Gamma_s, \quad c_s \in \mathbb{R}, \quad \sum_{s=1}^m c_s^2 = 1, \quad (2)$$

where the Γ_s are distinct nonidentity Pauli strings satisfying $\Gamma_s \Gamma_t = -\Gamma_t \Gamma_s$ whenever $s \neq t$. Such observables arise naturally from Clifford generators and fermion-to-qubit encodings.

3 Proof intuition

The unitarity of (2) is immediate: all cross terms cancel in A^2 . The influence problem becomes a weighted incidence problem. Give Γ_s weight $w_s = c_s^2$. Every pair of strings must locally anticommute on at least one qubit. At a fixed qubit, pairs locally anticommute only across the three Pauli labels X, Y, Z ; if their three total weights are x, y, z , the covered pair weight is $xy + xz + yz \leq (x + y + z)^2/3$. Thus n qubits can cover all pairs only if some qubit carries substantial total weight. A single dominant coefficient is handled separately by observing that one qubit in its support already has at least that much influence. Balancing these two alternatives yields an explicit $\Theta(n^{-1/2})$ bound.

The argument does not extend automatically to a general quantum Boolean function. In a general Pauli expansion, commuting pairs may have identical products with cancelling signed coefficients in $A^2 = I$. Controlling that cancellation geometry is precisely what remains open.

4 The partial theorem

Theorem 1 (Anticommuting-Pauli quantum KKL). *Let A be an anticommuting-Pauli observable on n qubits. Then A is a balanced quantum Boolean function and*

$$\max_{1 \leq j \leq n} \text{Inf}_j^2(A) \geq \rho_n := \frac{\sqrt{24n+9}-3}{4n} = \frac{6}{\sqrt{24n+9}+3}. \quad (3)$$

Consequently,

$$\max_j \text{Inf}_j^2(A) \geq \frac{6}{\sqrt{33}+3} \frac{1}{\sqrt{n}} \geq \frac{3e}{\sqrt{33}+3} \frac{\log n}{n}. \quad (4)$$

In particular, the Montanaro–Osborne conjecture holds for this class, with the stronger order $n^{-1/2}$.

Proof. Hermiticity is clear. Since $\Gamma_s^2 = I$ and distinct terms anticommute,

$$A^2 = \sum_s c_s^2 I + \sum_{s < t} c_s c_t (\Gamma_s \Gamma_t + \Gamma_t \Gamma_s) = I.$$

Every Γ_s is nonidentity and hence traceless, so $\text{tr}(A) = 0$.

Put $w_s = c_s^2$, so $w_s \geq 0$ and $\sum_s w_s = 1$. For each qubit j and $\alpha \in \{X, Y, Z\}$, define

$$u_{j,\alpha} = \sum_{s: (\Gamma_s)_j = \alpha} w_s, \quad q_j = u_{j,X} + u_{j,Y} + u_{j,Z}.$$

By (1), $q_j = \text{Inf}_j^2(A)$. Set $Q = \max_j q_j$.

Two Pauli strings anticommute exactly when the number of coordinates at which they carry two distinct nonidentity Pauli labels is odd. Therefore each unordered pair $s < t$ is locally anticommuting on at least one qubit. Counting the pair weight $w_s w_t$ at every such qubit gives

$$\begin{aligned} \sum_{s < t} w_s w_t &\leq \sum_{j=1}^n (u_{j,X} u_{j,Y} + u_{j,X} u_{j,Z} + u_{j,Y} u_{j,Z}) \\ &\leq \frac{1}{3} \sum_{j=1}^n q_j^2 \leq \frac{n}{3} Q^2. \end{aligned} \tag{5}$$

The middle inequality follows from $xy + xz + yz = ((x+y+z)^2 - (x^2 + y^2 + z^2))/2 \leq (x+y+z)^2/3$.

Let $a = \max_s w_s$. Since a string attaining a is nonidentity, some qubit lies in its support, and hence $Q \geq a$. Also $\sum_s w_s^2 \leq a \sum_s w_s = a$. Thus (5) and

$$\sum_{s < t} w_s w_t = \frac{1 - \sum_s w_s^2}{2}$$

imply

$$Q \geq \max \left\{ a, \sqrt{\frac{3(1-a)}{2n}} \right\}.$$

The first branch increases in a and the second decreases. Their intersection solves $a^2 = 3(1-a)/(2n)$, whose positive root is

$$a = \frac{\sqrt{24n+9} - 3}{4n} = \frac{6}{\sqrt{24n+9} + 3} = \rho_n.$$

This proves (3).

For $n \geq 1$, $\sqrt{24n+9} + 3 \leq (\sqrt{33} + 3)\sqrt{n}$, giving the first inequality in (4). Finally, the maximum of $(\log x)/\sqrt{x}$ on $x \geq 1$ is $2/e$, so $n^{-1/2} \geq e(\log n)/(2n)$. This proves the second inequality. $\square$

Corollary 2 (Equal-coefficient incidence form). *If $\Gamma_1, \dots, \Gamma_m$ are distinct nonidentity pairwise anticommuting Pauli strings and $A = m^{-1/2} \sum_{s=1}^m \Gamma_s$, then some qubit belongs to at least $m\rho_n$ of their supports.*

Proof. For this A , equation (1) says that the influence of qubit j is exactly the fraction of strings whose support contains j . Apply Theorem 1. $\square$

5 A current construction stress test

Slote's August 2026 Parseval-limit construction [4] produces homogeneous unitary Pauli expansions with exponentially many equal coefficients. It begins with an equal-weight, pairwise-anticommuting core $P_1, \dots, P_L$, then attaches many commuting diagonal suffix terms. This makes it a natural possible counterexample mechanism.

It does not evade the theorem. Each core index ℓ carries total squared Fourier mass $1/L$, so the influence of a core qubit is exactly its weighted incidence among the P_ℓ . Applying Theorem 1 to the normalized core gives a core influence of order at least the inverse square root of the number of core qubits. This is much larger than the KKL scale for the full construction. Thus the construction refutes quantum Fourier entropy-influence but not quantum KKL.

6 Verification, novelty, and limitations

The script `code/verify_small_cases.py` exhausts every subset of the 15 nonidentity two-qubit Pauli strings, retains the pairwise-anticommuting families, and checks the theorem and the intermediate pair-cover inequality for deterministic weight samples. It also checks random greedy families on three qubits. No violation was found. This is a finite consistency check and is not used in the proof.

A bounded novelty search was run through 9 August 2026. The sources checked included arXiv:0810.2435, 2209.07279, 2411.12399, and 2608.01424, plus exact and close web searches for “pairwise anticommuting Pauli influence KKL”, “anticommuting Pauli quantum KKL”, and “Clifford quantum KKL influence”. The searches found standard uses of anticommuting Pauli families and the known $p < 2$ quantum KKL variants, but not the weighted incidence theorem above. Novelty confidence is moderate, not definitive.

The full L^2 conjecture remains open. The theorem covers only Pauli expansions whose entire nonzero support is pairwise anticommuting. It does not control general unitaries with cancellations among products of commuting Pauli terms. The exponent $1/2$ is a universal lower bound for this class; no claim of sharpness of the exact constant is made.

Human-review recommendation. Review as a likely-valid substantial partial result. The highest-value checks are the inequality direction in the weighted pair cover (5), the Fourier normalization in (1), and the one-variable optimization producing ρ_n .

References

- [1] A. Montanaro and T. J. Osborne, *Quantum Boolean functions*, Chicago Journal of Theoretical Computer Science (2010), arXiv:0810.2435.
- [2] C. Rouzé, M. Wirth, and H. Zhang, *Quantum Talagrand, KKL and Friedgut’s theorems and the learnability of quantum Boolean functions*, Communications in Mathematical Physics **405** (2024), Article 95, arXiv:2209.07279.
- [3] Y. Jiao, W. Lin, S. Luo, and D. Zhou, *Quantum KKL-type inequalities revisited*, arXiv:2411.12399 (2024).
- [4] J. Sloate, *Dense Hamiltonians at the Parseval limit: The noncommutative BH constant is exponential and the quantum FEI conjecture is false*, arXiv:2608.01424 (2026).